

Math 170A: Probability Theory

Fall 2018

Instructor:	Elizaveta (Liza) Rebrova
E-mail:	rebrova@math.ucla.edu (please, put "170A" in the subject line of your email)
Course Webpage:	https://ccle.ucla.edu/course/view/18F-MATH170A-1
Instructor Webpage:	http://www.math.ucla.edu/~rebrova/170A.html (homework will be here)
Lectures:	
Room	5137 MS
Time	MWF 10:00am – 10:50am
Office Hours: (tentative)	
Room	5338 MS
Hours	MF 11:00am – 12:30pm (or by appointment)
Teaching Assistant	Robbie Housden
Discussion session room	5137 MS
Discussion session hours	R 10:00am – 10:50am
Office hours room	3915B MS
Office hours time	TBA

Course description

Probability and stochastic processes are used to create and analyze models in a broad range of fields, including statistics, economics, finance, engineering, biology and physics. Mathematics 170AB and 171 are designed to give a firm foundation in this area for students who will work and/or do graduate work in one of these fields. 170A is the first class in the sequence and is a prerequisite to 170B and 171.

Course materials

- **Textbook:** Dimitri P. Bertsekas and John N. Tsitsiklis *Introduction to Probability*, 2nd edition

Grading

Your grade is formed based on seven homework assignments, two midterms, and the final exam. MyUCLA gradebook will be used for grading. The final score will be obtained as the maximum of the following two schemes:

- homeworks - 15%, the 1st midterm - 20%, the 2nd midterm - 20%, the final - 45%
- homeworks - 15%, the largest of the midterm grades - 30%, the final - 55%

Any grading complaints and requests for re-grading must be submitted **within 1 week of receiving your score**. Grading complaints not initiated within this period of time will not be considered.

Homework

- Homework assignments will be issued weekly and posted on the instructor website. It is turned in the beginning of each discussion section (and returned on the next discussion section).
- Your score on homework assignment will be based on a few randomly chosen problems, as well as overall completion of the assignment.
- **Late homework will not be accepted.**
- One lowest homework score will be dropped.
- Printed or handwritten homeworks have to be submitted.
- You may use whatever resources you want to do the homework, including computers, textbooks, friends, the TA, etc. Collaboration in small groups (up to 5 people) is encouraged. You can indicate the names of your collaborators if you'd like to. However, **all homework assignments must be written by you, i.e. you cannot copy someone else's solution or submit a joint solution made by several people.**

Exams

- There will be two midterms and one final exam.

1st midterm: October 19 (Friday), 10:00am – 10:50am

2nd midterm: November 16 (Friday), 10:00am – 10:50am

Final: December 10 (Monday), 3pm-6pm

- You **must take the final exam** in order to pass the class.
- **No make-up exams will be possible.**
- Students must bring their UCLA ID cards to the midterms and to the final exam.

Tentative course schedule

This is a *very* tentative schedule:

Week	Date	Section	Topics
0	M		No class
	W		No class
	R	Discussion	Intro
	F, Sep 28	1.1	Sets and manipulating sets
1	M, Oct 1	1.2	Sample spaces & Probability laws
	W, Oct 3	1.3	Conditional Probability
	R	Discussion	Discussion
	F, Oct 5	1.3	Conditional Probability
2	M, Oct 8	1.4	Total Probability Theorem and Bayes Rule
	W, Oct 10	1.5	Independence
	R	Discussion	HW collected
	F, Oct 12	1.5	Independence
3	M, Oct 15	1.6	Counting
	W, Oct 17	2.1	Discrete Random Variables & Review
	R	Discussion	HW collected & Review
	F, Oct 19	1.1-2.1	Midterm 1
4	M, Oct 22	2.2	Probability Mass Functions
	W, Oct 24	2.3	Functions of Random Variables
	R	Discussion	Midterm discussion
	F, Oct 26	2.4	Expectation and Variance
5	M, Oct 29	2.5, 4.2	Joint PMFs of Multiple Random Variables
	W, Oct 31	2.6	Conditioning
	R	Discussion	HW collected
	F, Nov 2	2.6	Conditioning
6	M, Nov 5	2.7	Independence
	W, Nov 7	3.1	Continuous Random Variables and PDFs
	R	Discussion	HW collected
	F, Nov 9	3.2	Cumulative Distribution Functions
7	M, Nov 12	No class	Veterans day
	W, Nov 14	1.1-3.2	Review
	R	Discussion	HW collected
	F, Nov 16	1.1-3.2	Midterm 2
8	M, Nov 19	3.3	Normal Random Variables
	W, Nov 21	3.4	Joint PDFs of Multiple Random Variables
	R	No class	Thanksgiving
	F, Nov 23	No class	Thanksgiving
9	M, Nov 26	3.5	Conditioning
	W, Nov 28	3.6	The Continuous Bayes Rule
	R	Discussion	HW collected
	F, Nov 30	5.1	Weak Law of Large Numbers
10	M, Dec 3	5.2	Weak Law of Large Numbers
	W, Dec 5		Central Limit Theorem
	R	Discussion	HW collected
	F, Dec 7	all	Review

Notice about academic integrity

From the office of the Dean of Students:

“With its status as a world-class research institution, it is critical that the University uphold the highest standards of integrity both inside and outside the classroom. As a student and member of the UCLA community, you are expected to demonstrate integrity in all of your academic endeavors. Accordingly, when accusations of academic dishonesty occur, The Office of the Dean of Students is charged with investigating and adjudicating suspected violations. Academic dishonesty includes, but is not limited to, cheating, fabrication, plagiarism, multiple submissions or facilitating academic misconduct.”

Students are expected to be aware of the University policy on academic integrity in the UCLA Student Conduct Code:

http://www.deanofstudents.ucla.edu/Portals/16/Documents/UCLACodeOfConduct_Rev030416.pdf

Please note the sections on (1) cheating, (2) plagiarism, and (3) unauthorized study aids.

Violation of course policy involving plagiarism, cheating, or possession of course materials during exams will be referred to the Dean of Students, who will be encouraged to take strong action. Do not cheat! The penalties can be very harsh. Do not believe it if you hear that “everyone does it.” You generally do not hear about the punishments because they are kept confidential. If you are found responsible by the Dean of Students for violating course policy, cheating on any course materials, or giving or receiving unauthorized help, a zero will be assigned for the entire assignment. No exceptions will be made! Past examples of penalties also include loss of an entire term of credit and suspension for several terms. If you plan to apply to graduate or professional school, such a negative mark on your record may be a major obstacle to admission.

No cell phones are allowed during exams. They must be left in your bag and turned off, or submitted to the designated TA/proctor. Students may not use a cell phone as a clock to keep time, nor as a calculator. No hats are allowed in the testing room; they must be left in your bag.

Notice about sexual harassment, discrimination, and assault

Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking. Students who have experienced sexual harassment or sexual violence can receive **confidential** support and advocacy from a CARE advocate:

The CARE Advocacy Office for Sexual and Gender-Based Violence

1st Floor, Wooden Center West

CAREadvocate@caps.ucla.edu

(310) 206-2465

You can also report sexual violence or sexual harassment directly to the University’s Title IX Coordinator:

Kathleen Salvaty

2241 Murphy Hall

titleix@conet.ucla.edu

(310) 206-3417